

Quantum Chemistry -- Molecular Orbitals

Subject: Quantum Chemistry | Topic: Molecular Orbitals & Born-Oppenheimer Approximation

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1. The Molecular Hamiltonian

A full molecular Hamiltonian includes electronic kinetic energy, nuclear kinetic energy, nuclear-nuclear Coulomb repulsion, electronic-nuclear Coulomb attraction, and electron-electron Coulomb repulsion terms.

2. The Born-Oppenheimer Approximation

Because atomic nuclei are vastly more massive than electrons ($m_p / m_e > 1836$), the velocities of electrons are orders of magnitude greater than nuclear velocities.

Under the Born-Oppenheimer approximation, nuclear positions are treated as clamped parameters while solving for electronic states.

3. Linear Combination of Atomic Orbitals (LCAO)

Molecular orbitals ψ_{MO} are approximated as linear combinations of localized atomic orbitals: $\psi = \sum_i c_i \phi_i$. Solving the secular equation produces bonding orbitals (lower energy, constructive interference) and antibonding orbitals.

4. Potential Energy Curves

Plotting the electronic eigenvalue plus nuclear repulsion as a function of internuclear distance yields adiabatic potential curves describing molecular bond stability and vibration.